

DATA AND MODEL FLOW

VERIFICATION AND EVIDENCE

Retained waveform records

HDF5 trace names; E/N/Z; arrival labels

Numerical identity

2,234 unique matches in local HDF5
Recovered source mapping; CSV byte hashes

Source partition and augmentation

Train 1,544 / validation 360 / test 330
Augmented records retain parent source

Membership and parentage

Zero source intersections across outer splits
Assert parent identity after augmentation

Grouped bases and OOF features

Five source folds; separate inner validation
Fold bases supply held-source features

OOF and feature contracts

Held sources excluded from fit / inner val
Assert producers, order and 15-channel schema

Meta-learner and saved models

15 channels; three base × three meta seeds
Outer validation reused for final selection

Artifact provenance

106 checkpoint hashes and memberships
Frozen manifests link inputs and models

Decoder and fixed-split evaluation

Frozen settings; 70 test prediction sets
Source-cluster intervals; no retuning

Replay and metric audit

18,528 OOF + 1,980 test tensors
70 prediction sets; 1,260 metric rows

Historical development exposure remains: these checks do not establish independent confirmation.

Solid vertical arrows: computational flow. Horizontal arrows: checks applied to each stage.